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An embedded system for SIDS prevention

Newborns Signals Monitoring (NICU)

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ABSTRACT

This study presents the design and conceptual development of a low-cost, home-based infant monitoring device aimed at early detection of physiological abnormalities and apnea events. The proposed system continuously monitors key vital parameters including heart rate (HR), oxygen saturation (SpO_2), body temperature, estimated blood pressure (BP), and apnea episodes using non-invasive sensing techniques. Heart rate is obtained using a dual sensing approach, combining electrocardiography (ECG) electrodes and photoplethysmography (PPG) to improve measurement reliability. SpO_2 is measured using PPG, while apnea detection is performed through analysis of oxygen desaturation patterns and signal interruptions associated with breathing cessation. Body temperature is monitored using a skin-contact temperature sensor, and cuffless BP estimation is explored using pulse timing features. The system is designed specifically for home monitoring, with emphasis on low cost, comfort, safety, and minimal skin irritation suitable for infant use. Wireless communication enables real-time data visualization, data logging, and alert generation for caregivers when abnormal physiological conditions are detected. The proposed device is intended to help caregivers monitor infants at home and identify potential health issues early, while not being designed to replace professional medical diagnosis..

Keywords: *Infant monitoring, Low-cost medical device, ECG, PPG, Apnea detection.*

1. INTRODUCTION

According to UNICEF, approximately **2.3** million children worldwide died within the first month of life in **2023**, corresponding to nearly **6,300** neonatal deaths each day. Many of these deaths occur among preterm and low birth weight infants, Infants born prematurely or with low birth weight have significantly higher risks of neonatal morbidity and mortality due to immature organ development and increased vulnerability to external infections and environmental factors [1,2]. Continuous monitoring of vital signs plays an important role in the early identification of physiological abnormalities in these high-risk infants.

Monitoring vital signs in infants is particularly important, as newborns are more sensitive to sudden changes in oxygen levels, heart activity, and body temperature. In home environments, such changes may not always be noticed immediately by caregivers. Early detection of conditions such as oxygen desaturation, abnormal heart rate, fever, or apnea can help caregivers take timely action and seek medical support when needed.

With advances in biomedical sensing and wearable technologies, several infant monitoring systems have been developed. However, many existing solutions are designed for hospital use, are expensive, or may be uncomfortable for long-term application on infant skin. Therefore, this project focuses on the development of a low-cost, home-based infant monitoring device that aims to monitor heart rate, oxygen saturation, body temperature, estimated blood pressure, and apnea events using non-invasive methods.

1.1 Literature Survey

Previous studies have emphasized the importance of continuous monitoring of vital signs in neonatal and infant care, particularly for early detection of physiological deterioration and life-threatening conditions. Neonatal mortality and severe morbidity are most likely to occur during the early postnatal period, highlighting the need for effective monitoring systems during this critical time [1]. These findings underline the importance of timely detection of abnormalities such as oxygen desaturation, abnormal heart activity, and apnea.

Noninvasive monitoring techniques have gained significant attention in recent years due to their safety and suitability for neonatal applications. Photoplethysmography (PPG) has

been widely used for measuring heart rate and oxygen saturation, as it enables continuous monitoring without invasive procedures [3]. In addition, electrocardiography (ECG) remains a reliable method for heart rate detection and is often used as a complementary technique to improve measurement accuracy and signal reliability. Furthermore, several studies have explored **cuffless blood pressure estimation** using the **pulse arrival time (PAT) method**, which is based on the time delay between the electrical activity of the heart measured by ECG and the peripheral pulse detected by PPG. The PAT-based approach offers a promising noninvasive alternative for blood pressure estimation, particularly in neonatal and infant monitoring, where traditional cuff-based measurements may be uncomfortable or impractical [4].

Several studies have investigated the use of wearable and home-based monitoring devices for infants. While many systems are designed for clinical or intensive care environments, fewer solutions focus on low-cost home monitoring applications. Existing commercial devices may be limited by high cost, reduced comfort, or susceptibility to motion artifacts, which can affect signal quality and reliability [5].

Based on the reviewed literature, there is a clear need for a low-cost, noninvasive infant monitoring system that combines multiple sensing methods to improve reliability while maintaining comfort and safety. This project builds upon previous research by integrating ECG and PPG for heart rate monitoring, along with SpO₂, temperature, estimated blood pressure, and apnea detection, aiming to support caregivers in home environments.

1.2 Project Goals and Objectives

The main goal of this graduation project is to design and conceptually develop a low-cost, non-invasive infant monitoring system that can be used in home environments, particularly for infants who require continuous observation after discharge from the Neonatal Intensive Care Unit (NICU). The proposed system aims to support caregivers by providing continuous monitoring of essential physiological parameters and early detection of potential health abnormalities.

The primary objective of this project is to monitor key vital signs, including heart rate, oxygen saturation (SpO₂), body temperature, estimated blood pressure, and apnea events, using safe and non-invasive sensing techniques suitable for infant use. Heart rate monitoring is intended to be implemented using a dual-sensing approach that combines

electrocardiography (ECG) and photoplethysmography (PPG) to improve signal reliability and reduce the effect of motion artifacts [6,7,8]. Oxygen saturation measurement and apnea detection are planned to be based on PPG signal analysis and oxygen desaturation patterns [9].

Another objective of this project is to explore cuffless blood pressure estimation using pulse timing features derived from ECG and PPG signals. While this method does not aim to provide clinical-grade blood pressure values, it is intended to offer an estimated trend that may help identify abnormal physiological changes. Body temperature monitoring is also included to allow early detection of fever or hypothermia, which are critical indicators in infant health monitoring [10,11].

In addition to physiological monitoring, this project aims to design a wireless data transmission system that enables real-time data visualization, data logging, and alert generation. Alerts are intended to notify caregivers when measured parameters exceed predefined safe thresholds. The system is designed with an emphasis on low cost, ease of use, comfort, and safety, ensuring minimal skin irritation and suitability for long-term monitoring.

Overall, the objective of this graduation thesis is not to replace professional medical equipment or diagnosis, but to provide a supportive monitoring tool that can assist caregivers in observing infants at home and identifying potential health issues at an early stage.

2. MATERIALS AND METHODS

Figure 1 illustrates the general process flow of the proposed infant monitoring system, starting from signal acquisition and continuing through signal processing, parameter estimation, and data transmission to the caregiver interface.

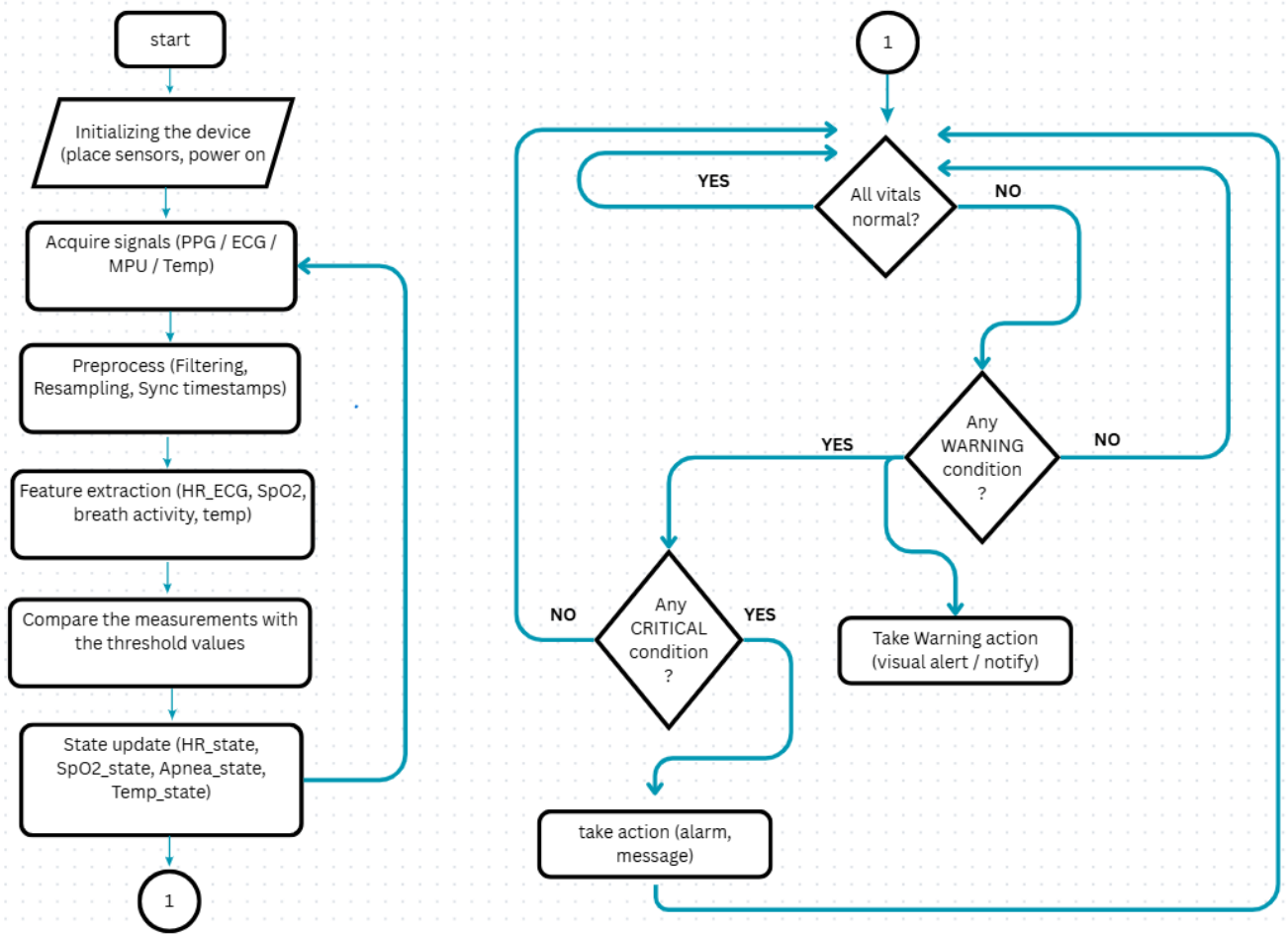


Figure 1. Flowchart illustrating the decision-making and state classification process of the monitoring system.

2.1 Materials

The main hardware components used in the proposed infant monitoring system and their corresponding functions are summarized in **Table 1**. These components were selected based on their non-invasive nature, suitability for infant monitoring, availability, and cost considerations.

Table 1 shows the list of sensors, processing units, and electronic components used in the system, along with their primary functions. An ECG sensor module (AD8232) was used to acquire cardiac electrical signals for heart rate estimation. A PPG-based sensor (MAX30102) was used to measure heart rate and oxygen saturation, serving as a backup and complementary method to ECG-based measurements. An inertial measurement unit (IMU), specifically the MPU6050 accelerometer, was included to support respiration-related motion detection and to assist in identifying apnea events during sleep. Body temperature measurement was performed using the MAX30205 temperature sensor, which is suitable for skin-contact applications [12].

An ESP32 module was used for data acquisition and wireless transmission, while an STM32 microcontroller was considered for additional data processing and system control. Together, these components formed the hardware foundation of the proposed monitoring device, as summarized in **Table 1**.

Component name	Function
AD8232 ECG Sensor	Estimate the heart rate through ECG signal
MAX30102 sensor	Measure spo ₂ and heart rate (back up for the heart rate)
IMU/accelerometer (MPU6050).	Respiration rate (especially for detecting sleeping apnea)
ECG electrodes with help of PPG	derive information about blood pressure (BP).
MAX30205	Measures the temperature of the body
Esp32	Receives data from all sensors then send them to the controller device
Microcontroller (STM32)	Receives all data sent by the ESP32, then it will process them, and send any alarms or notification.

Table 1. Hardware components used in the proposed infant monitoring system and their functions.

In addition to the hardware components described previously, this study also defined a set of derived physiological signals and system states that were used for monitoring, analysis, and decision-making. These derived variables were obtained through signal processing and feature extraction from the acquired sensor data.

The derived physiological parameters and system states used in the proposed monitoring system are summarized in **Table 2**. As shown in Table 2, the final heart rate (HR) was selected by combining and comparing measurements obtained from ECG and PPG signals to improve reliability. Oxygen saturation (SpO₂) was derived from the PPG signal and expressed as a percentage. The apnea state variable was used to classify breathing status into three categories: none, suspected, or confirmed apnea. Body temperature was recorded in degrees Celsius, and a system risk level was defined to classify the overall condition of the infant as normal, warning, or critical based on the measured parameters. These derived variables were used by the system to evaluate the infant's physiological condition continuously and to support alert generation when abnormal values were detected.

Variable	Description	Unit
HR	Final heart rate selected from ECG and PPG	bpm
SpO ₂	Oxygen saturation	%
Apnea state	Apnea detection state (None / Suspected / Confirmed)	–
Temp	Body temperature	°C
Risk level	System risk classification (Normal – Warning - Critical)	-

Table 2. Derived physiological signals and system states used in the proposed monitoring system.

To support data storage, visualization, and analysis, a structured dataset format was also defined. The dataset fields used in this study are presented in **Table 3**. As reported in Table 3, each data record included a timestamp to indicate the time of measurement, heart rate values obtained separately from ECG and PPG signals, the selected final heart rate, oxygen saturation, body temperature, and the system risk level. This dataset structure allowed continuous logging of physiological data and enabled retrospective

analysis of trends and abnormal events.

The defined dataset fields were selected to ensure clarity, consistency, and ease of interpretation for caregivers and researchers. These materials formed the basis for system monitoring, decision-making, and alert generation throughout the study.

Field name	Description
timestamp	Time of measurement
HR_ECG	Heart rate from ECG
HR_PPG	Heart rate from PPG
HR	Selected heart rate
SpO ₂	Oxygen saturation
Temp	Body temperature
BP	Blood Pressure
Risk level	Alarm severity

Table 3. Dataset fields used for data logging and analysis.

2.2 Methods

The operational workflow of the proposed infant monitoring system is illustrated in **Figure 1**, which presents the step-by-step process followed during signal acquisition, processing, and decision-making.

As shown in **Figure 1**, the system operation started with device initialization, which included powering the system and placing the sensors on the infant's body. After initialization, physiological signals such as ECG, PPG, motion data from the IMU, and body temperature were acquired continuously. These raw signals were then preprocessed using filtering, resampling, and time synchronization techniques to reduce noise and improve signal quality.

Following preprocessing, feature extraction was performed to obtain vital parameters including heart rate, oxygen saturation (SpO₂), respiration-related activity, and body temperature. The extracted parameters were then compared with predefined threshold values to assess the infant's physiological state.

As indicated in **Figure 1**, if all measured vital signs were within normal ranges, the system continued monitoring without generating alerts. If abnormal values were detected, the system evaluated whether the condition corresponded to a warning or a critical state. Warning conditions triggered visual alerts or notifications to caregivers, while critical conditions resulted in immediate alarm generation and emergency notifications.

This flowchart-based approach ensured continuous monitoring, timely detection of abnormalities, and appropriate alert generation, supporting early intervention while maintaining simplicity and reliability in system operation.

3. RESULTS AND DISCUSSION

In this section, the results obtained from the implementation and testing of the proposed infant monitoring system are presented and discussed. The results are based on recorded physiological data, system state classification, and hardware-level testing. Tables and figures are used to demonstrate system performance and to support the discussion.

3.1 Recorded Physiological Data and System States

During system testing, physiological parameters including heart rate, oxygen saturation, body temperature, estimated pressure, and system states were recorded continuously and stored in a structured dataset. A sample of the recorded dataset is presented in **Table 4**, which includes timestamped measurements and corresponding system classifications.

As shown in **Table 4**, the estimated heart rate values varied over time, reflecting normal fluctuations during the resting state. Oxygen saturation values remained close to the expected physiological range, while body temperature values were recorded continuously using the skin-contact temperature sensor. Based on these measurements, the system generated individual parameter states (HR_state, SpO₂_state, TEMP_state) and an overall system state.

In several instances, the system classified the overall state as *critical*, primarily due to abnormal temperature readings. This demonstrates that the threshold-based decision logic was able to detect abnormal conditions and escalate system alerts accordingly. The results confirm that the system was capable of combining multiple physiological parameters to generate an overall risk classification rather than relying on a single measurement.

t_ms	hr_est	spo2_ratio	tempC	press_hPa	segment	HR_state	SPO2_state	TEMP_state	state
20000	105.5	1.001	20.55	999.34	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
21000	97.6	1.003	20.6	999.37	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
22000	98.6	0.995	20.66	999.29	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
23000	83.9	0.994	20.65	999.35	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
24000	93.9	1.001	20.62	999.36	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
25000	94.1	1.012	20.58	999.34	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
26000	88.8	0.999	20.61	999.37	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
27000	99.9	1.001	20.61	999.3	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
28000	100.8	1	20.62	999.35	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
29000	88.3	1.002	20.61	999.36	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
30000	97.3	0.996	20.59	999.27	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
31000	96.4	1	20.62	999.33	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
32000	85.2	1	20.6	999.34	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
33000	82.3	0.999	20.61	999.33	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
34000	102.5	0.996	20.63	999.37	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
35000	94.9	1	20.6	999.35	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
36000	114.5	1.004	20.65	999.29	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
37000	95.6	0.992	20.62	999.33	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
38000	88.8	1.001	20.6	999.33	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
39000	92	1.009	20.62	999.37	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
40000	98.9	1.005	20.6	999.38	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
41000	108.3	0.996	20.63	999.39	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
42000	97	0.997	20.6	999.33	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
43000	96.2	0.997	20.61	999.3	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL
44000	63.2	0.993	20.62	999.35	REST	NORMAL	NORMAL	CRITICAL	CRITICAL
45000	91.1	1.003	20.6	999.32	REST	CRITICAL	NORMAL	CRITICAL	CRITICAL

Table 4. Sample recorded physiological data and corresponding system state classifications.

3.2 ECG Electrode Placement and Signal Acquisition

Electrocardiogram (ECG) signals were acquired using a three-electrode configuration, as illustrated in **Figure 2**. The electrodes were placed on the right arm (RA), left arm (LA), and right leg (RL) positions, which represent a standard and widely used configuration for single-lead ECG measurement.

As indicated in **Figure 2**, this electrode placement allowed reliable acquisition of cardiac electrical activity while maintaining simplicity and comfort. The selected configuration was suitable for experimental testing and aligns with common practices used in wearable and portable ECG systems. The obtained ECG signals were used for heart rate estimation and to support pulse timing-based analysis.

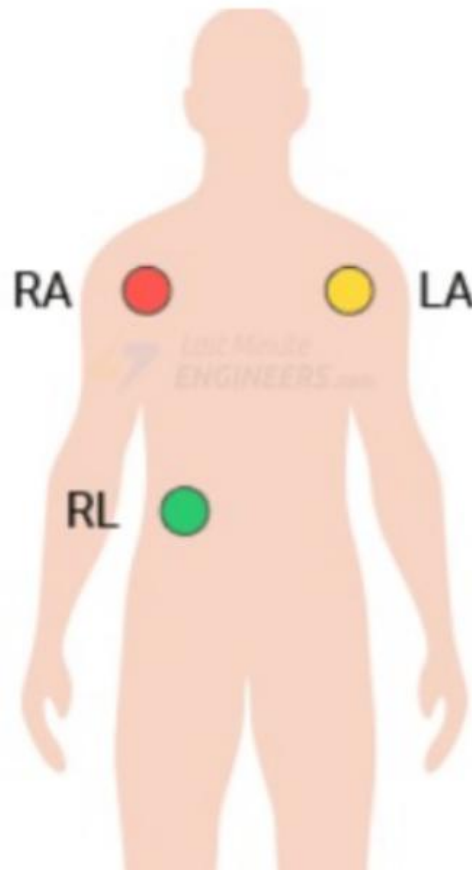


Figure 2. ECG electrode placement configuration used for heart rate signal acquisition

3.3 Hardware Prototype and System Implementation

The physical implementation of the proposed monitoring system is shown in **Figure 2**, which presents the hardware prototype used during testing. The prototype consisted of a microcontroller unit, sensor modules, connecting wires, and visual indicators mounted on a breadboard.

As shown in **Figure 3**, the system successfully interfaced with the temperature sensor and visual alert indicators. Light-emitting diodes (LEDs) were used to represent different system states and to provide immediate visual feedback when abnormal conditions were detected. This prototype setup demonstrated the feasibility of real-time data acquisition, processing, and alert generation at the hardware level.

Although the prototype was implemented using development boards and breadboard connections, it served as a functional proof of concept for the proposed system design.

The results indicate that the system architecture can be further miniaturized and optimized for wearable and home-based applications.



Figure 3.1 Complete monitoring system and mobile app.

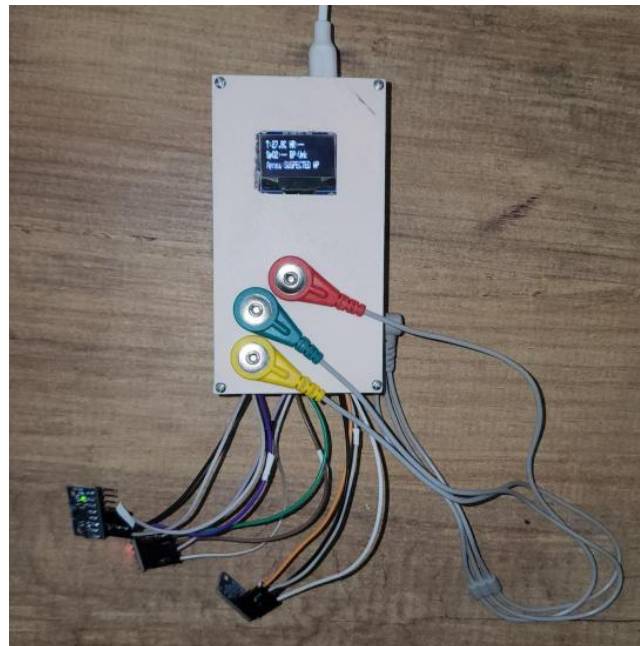


Figure 3.2 Hardware prototype with sensors and ECG leads.

3.4 Embedded System Implementation and Serial Output Results

The embedded firmware of the proposed monitoring system was implemented and tested using a microcontroller platform. The program handled sensor data acquisition, threshold comparison, state classification, and serial data transmission.

Figure 4 presents the Arduino IDE interface and real-time serial monitor output during system operation. The serial output shows timestamped measurements of heart rate, oxygen saturation, body temperature, estimated pressure, individual parameter states, and the overall system state. The system successfully classified physiological parameters based on predefined thresholds, generating **NORMAL** or **CRITICAL** states accordingly. These results confirm correct firmware execution, functional threshold logic, and reliable real-time data logging suitable for monitoring and offline analysis.

```
ECG HR: 82 BPM | SpO2: 96 % | Temp: 30.06 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 82 BPM | SpO2: 96 % | Temp: 30.06 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
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ECG HR: 83 BPM | SpO2: 96 % | Temp: 30.06 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 83 BPM | SpO2: 96 % | Temp: 30.06 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 83 BPM | SpO2: 96 % | Temp: 30.06 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 83 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 83 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 83 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 84 BPM | SpO2: 96 % | Temp: 30.06 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 86 BPM | SpO2: 96 % | Temp: 30.06 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 86 BPM | SpO2: 96 % | Temp: 30.06 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 88 BPM | SpO2: 96 % | Temp: 30.06 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 89 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 89 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
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ECG HR: 88 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 88 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 88 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 87 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 87 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 87 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
ECG HR: 87 BPM | SpO2: 96 % | Temp: 30.12 C | PAT: 51 ms | BP Trend: BP tendency decreasing | Estimated BP: 85/46 mmHg
-- Finger removed. Waiting... --
-- Finger detected. Measuring SpO2 and BP trend... --
-- Finger removed. Waiting... --
-- Finger detected. Measuring SpO2 and BP trend... --
```

Figure 4. Arduino IDE screenshot showing firmware implementation and real-time serial monitor output.

3.5 Discussion

The results obtained in this study demonstrate that the proposed infant monitoring system was able to acquire physiological signals, derive meaningful parameters, and classify system states based on predefined thresholds. The integration of multiple sensors allowed cross-verification of measurements and improved robustness against signal artifacts.

Compared with similar studies reported in the literature, the use of combined ECG and PPG signals for heart rate monitoring aligns with established approaches for improving reliability in non-invasive monitoring systems. Unlike clinical NICU systems, the proposed device prioritizes low cost, simplicity, and home usability rather than clinical-grade accuracy.

It is important to note that the results presented in this study represent system-level performance and feasibility testing rather than clinical validation. The recorded data were obtained under limited experimental conditions, and further testing involving infant subjects and comparison with clinical reference devices would be required for medical use.

3.6 Limitations

Despite the successful demonstration of system functionality, several limitations were identified in this study. The system was tested under controlled conditions and did not include long-term monitoring or clinical trials. Temperature measurements were influenced by environmental conditions and sensor placement, which may affect measurement stability. In addition, blood pressure estimation was limited to observing relative trends rather than providing absolute clinical values.

Another important limitation was related to the use of wired sensor connections. The presence of multiple cables increased system complexity and may reduce comfort,

particularly in long-term or infant monitoring scenarios. Wired connections also introduce a higher risk of motion artifacts, cable displacement, and handling difficulties, which can affect signal quality and overall system usability.

Future work should focus on extended validation under real-life conditions, improved sensor packaging, and enhanced motion artifact reduction techniques. In particular, transitioning to a fully wireless system should be prioritized to improve comfort, safety, and ease of use. The integration of wireless communication and compact sensor modules would make the system more suitable for continuous home monitoring and closer to practical NICU and post-NICU applications.

4. CONCLUSION

This study presented the design and preliminary implementation of a low-cost, non-invasive infant monitoring system intended for home use, particularly to support infants requiring continued observation after discharge from the Neonatal Intensive Care Unit (NICU). Motivated by the need for early detection of physiological abnormalities and the limitations of existing hospital-based monitoring systems in home environments, the proposed system integrated multiple sensors to monitor heart rate, oxygen saturation, body temperature, estimated blood pressure trends, and apnea-related events. The results demonstrated that the system could acquire physiological signals, deriving meaningful parameters, and classifying system states using threshold-based logic, thereby supporting continuous monitoring and early warning generation.

The proposed system was designed while considering realistic engineering constraints and real-life conditions. Non-invasive sensing techniques were prioritized to ensure infant safety and comfort, while low-cost components were selected to improve accessibility for home use. However, several limitations were identified. The system was evaluated under limited experimental conditions and did not undergo clinical validation. Sensor measurements were influenced by factors such as placement accuracy, motion artifacts, and environmental conditions. In addition, cuffless blood pressure estimation was limited to observing relative trends rather than providing absolute clinical values. These constraints highlight that the system is intended as a supportive monitoring tool rather than a replacement for professional medical equipment or diagnosis.

Despite these limitations, the proposed system demonstrates potential benefits for real-life applications, including increased caregiver awareness, continuous monitoring outside clinical settings, and early identification of abnormal physiological conditions. The modular system design allows future improvements, such as enhanced signal processing, improved sensor packaging, and compliance with medical engineering standards and regulations. Overall, this study provides a feasible foundation for low-cost infant monitoring and contributes to the ongoing development of accessible and safe healthcare technologies for neonatal and post-NICU care.

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